

	DOCUMENT TITLE: Suspension Quality Control Report	Revision: A
	DOCUMENT #: TMP1-QC-C.SLQ001	Effective Date: 23Sep2022

General Production Information

Work Order #:	N/A	NCMR #:	N/A
Part Number:	C.SLQ001	Manufacture & Expiration Date:	Mfg date: 25 Aug 2021 Exp date: 25 Aug 2022
Part Revision:	A	Batch / Lot #:	HYX-825021

Supplies Traceability

Part Number	Description	Part Revision	Lot / Batch #
S.SLQ005	pH Strip	A	10BDH4801
S.SLQ006	Deuterium oxide	A	M2338

Equipment Traceability

Part Number	Description	Equipment ID	Calibration Due Date
E.SLQ013	Weighing scale	ST-012	10/3/2023*

*Deviation and Justification: Equipment ST-012 (Weighing scale) was used in this quality control procedure passed its calibration due date. This QC procedure was executed on [date]. The weighing scale was sent out for recalibration on 02 Aug 2024 and was found to be in tolerance when received by the company performing the calibration service, Microprecision Calibration, Inc. Additionally, no adjustments were made to the weighing scale. Therefore, the data obtained using the scale for this QC procedure is deemed acceptable. The calibration report can be found in the equipment file for the weighing scale (ST-012).

QC Data

Sample Tube	Weight Empty (g)	Weight with Residue (g)	Residue Net Weight (g)
Sample A1	13.072	13.175	0.103
Sample A2	13.081	13.181	0.998
Sample B1	13.004	13.108	0.104
Sample B2	12.999	13.103	0.104
Sample C1	13.094	13.194	0.100
Sample C2	13.078	13.179	0.101

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Sample Tube	Weight Empty (g)	Weight with Sample suspended (g)	Weight with Residue on Evaporation (g)	Residue on Evaporation (%)
Suspension Test Sample	13.015	44.453		
Residue on Evaporation Sample	13.100	23.120	13.202	1.014%

Summary of Results

Attribute	Result	Acceptance criteria	Meets Criteria? (circle one)
Appearance (color)	Opaque White	Opaque white	☒ Yes ☐ No
Appearance (form)	Inviscid Liquid	Inviscid liquid	☒ Yes ☐ No
Residue on Evaporation	1.014%	0.5% – 1.5%	☒ Yes ☐ No
pH	6	5 – 8	☒ Yes ☐ No
¹ H NMR	Broad Peaks are present at approximately 0.99, 2.2, 2.9, 3.1, 3.5, 3.7, 4.4, and 4.75 ppm indicating successful incorporation of the zwitterion moiety into the polymer. See attached ¹ H NMR spectra.	Identify Peaks with Amplitude >5% of Solvent Peak (4.75ppm)	☒ Yes ☐ No
¹⁹ F NMR	Two peaks at -140 and -150 ppm identified. No other peak was present at -76ppm indicating there is no TFE remaining in the suspension. See attached ¹⁹ F NMR spectra.	Identify Peaks, if there is a peak at -76ppm determine approximate amplitude relative to peaks at -140ppm and -150ppm	☒ Yes ☐ No

Report completed by:

Ethan Rao
Print name

Ethan Rao
Signature

18 July 2024
Date

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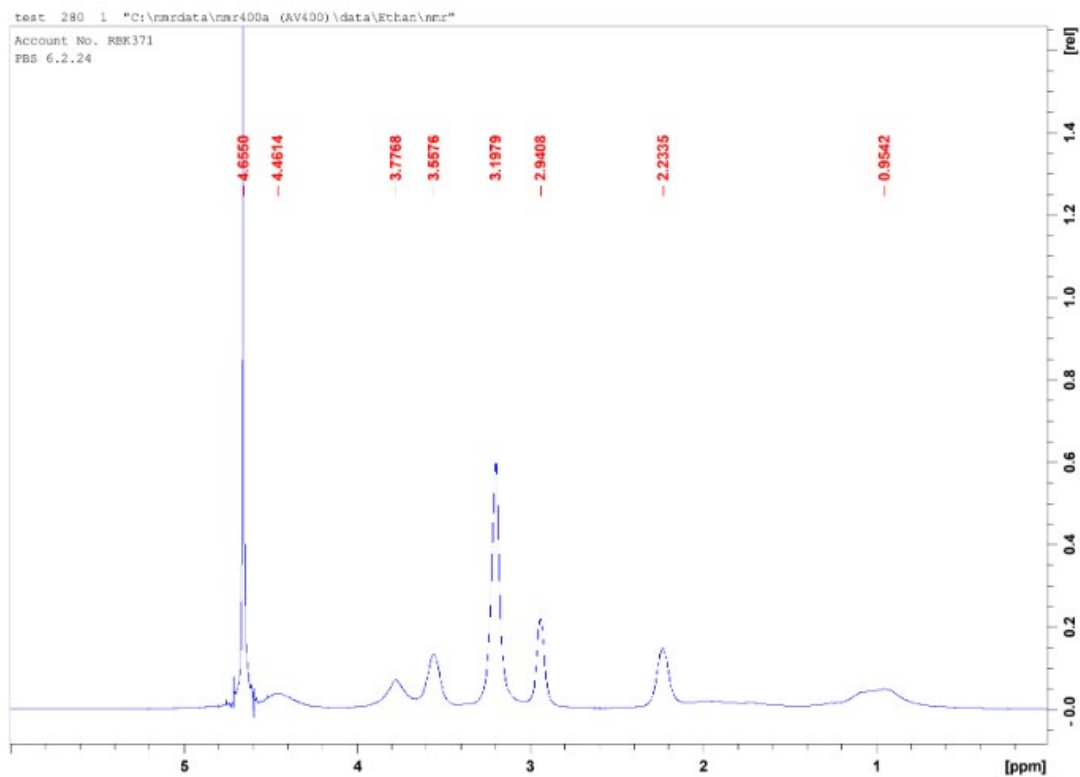
Report reviewed by:

Bryan Wong
Print name


Signature

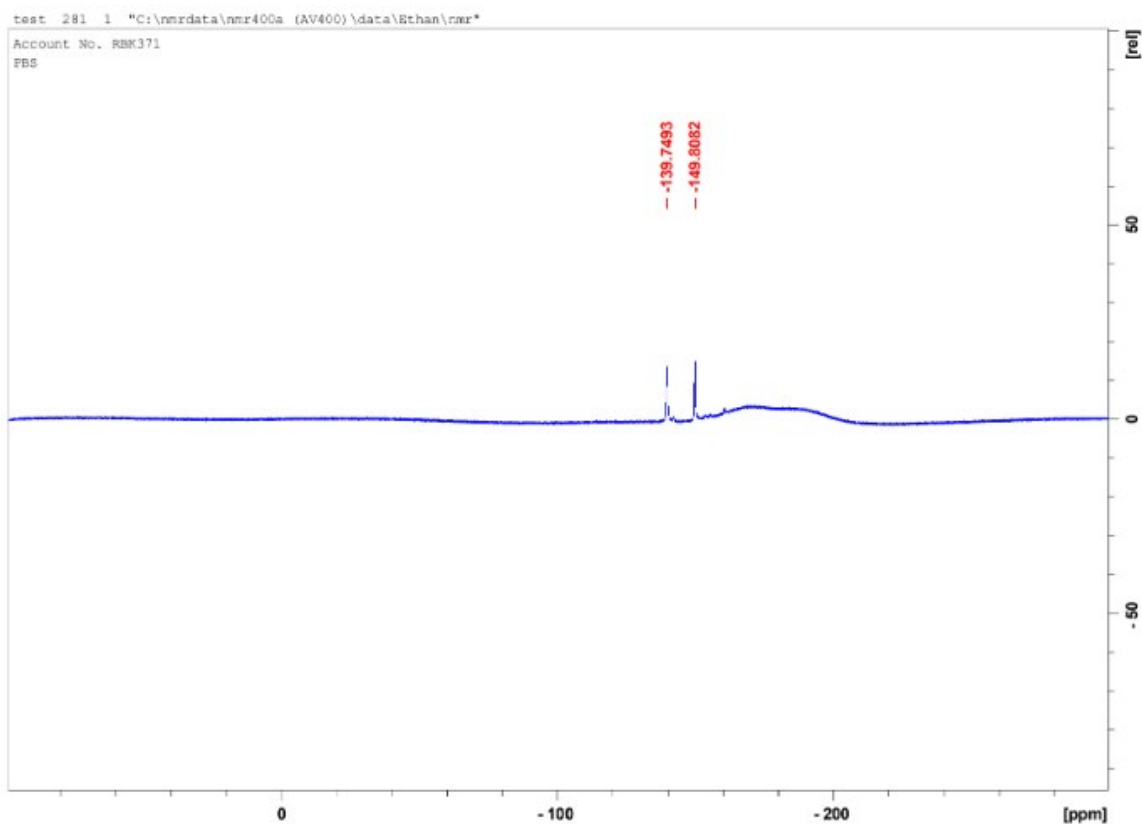
13 Nov 2024
Date

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^1H NMR: Broad Peaks are present at approximately 0.99, 2.2, 2.9, 3.1, 3.5, 3.7, 4.4, and 4.75 ppm indicating successful incorporation of the zwitterion moiety into the polymer.

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^{19}F NMR: Two peaks at -140 and -150 ppm identified. No other peak was present at -76ppm indicating there is no TFE remaining in the suspension.